

Sound Becomes a Signal

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can interpret an ordered sequence of sound-signal samples with a stated time interval.

I can explain why summaries or sparse samples can omit changes.

I can separate an audio recording, a transcript and an interpretation of meaning.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Computers receive changing measurements, not human-like hearing.



LEARNING TARGET A whisper recorded close to a microphone has larger sample amplitudes than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Learner name:

Readiness for the connected explanation

Two image grids have the same number of black pixels. Why can their pictures differ?

This week's end goal

By the end, I can connect ordered sound samples to timing and explain what a maximum loses.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a waveform.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

Sound involves changing pressure in a medium such as air. A microphone responds to that variation and produces a changing signal. A digital recording stores measurements of the signal at successive times. The values are ordered: swapping their times changes the represented pattern.

We will use tiny invented sequences to study this idea on paper. Their values are arbitrary signal units, not decibels or a test of safe listening levels. Real audio uses many more samples, so our five-point examples should not be played as if they were a realistic recording of speech.

A recording must include enough information to interpret the values, including their time spacing and encoding. The sequence 0,2,0,2,0 does not by itself say how quickly the changes happened. This is similar to needing image dimensions as well as pixel values.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



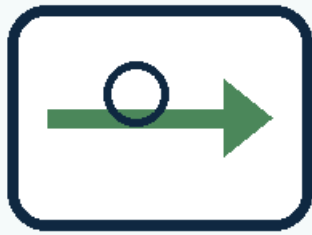
INPUT

changing air pressure



PROCESS

sample the signal and compare a clap
pattern



OUTPUT

clap detected, not detected, or uncertain

INPUT changing air pressure

PROCESS sample the signal and compare a clap pattern

OUTPUT clap detected, not detected, or uncertain

CORE IDEA Computers receive changing measurements, not human-like hearing.

UNDER THE HOOD Audio is represented by samples taken over time. Sample rate describes how often measurements are taken; amplitude describes signal strength. Noise can overlap the target pattern, so detection requires a decision rule or model and an uncertainty policy.

Draw a fresh example with all three stages:

A closer explanation

Three stages in a speech system

Input · Operation · Output

Input: Sound reaching a microphone

Operation: Measure the changing signal at successive times

Output: Ordered samples and timing information

Input: Recorded audio

Operation: Speech recognition estimates the words

Output: A transcript that may contain errors

Input: Transcript and an explicit request

Operation: A separate rule or model processes the text

Output: An answer or proposed action

Recording, transcribing and interpreting are separate operations. A later stage can correctly process an incorrectly transcribed number.

Time in milliseconds · 0 · 1 · 2 · 3 · 4

Time in milliseconds: Invented signal value

0: 0

1: 2

2: 0

3: 2

4: 0

There are five samples with four gaps between the first and last. Each gap is 1 millisecond, so the elapsed time from first to last is 4 milliseconds. If the same five values were spaced 2 milliseconds apart, they would span 8 milliseconds. Equal value lists with different timing do not describe the same time pattern.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

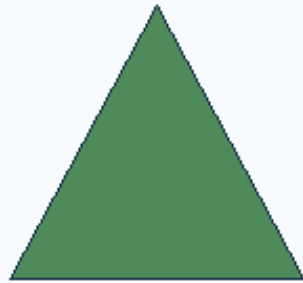
PICTURE GALLERY

Four Cases, Four Clues

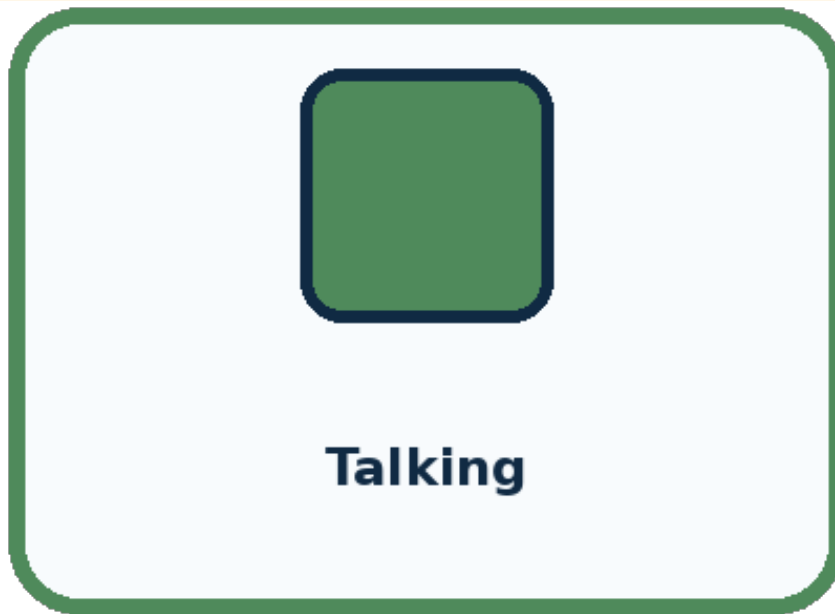
Use the visual cue, then read the evidence carefully.



One clap



Two claps



CASE 1 — One clap Clue: single sharp peak. Meaning: target pattern.

CASE 2 — Two claps Clue: two separated peaks. Meaning: different timing pattern.

CASE 3 — Talking Clue: many irregular changes. Meaning: possible false trigger.

CASE 4 — Quiet room Clue: small background values. Meaning: baseline.

BOUNDARY A microphone measures a signal; it does not understand a speaker's meaning by itself.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Retain only samples at 0,2,4 ms.

Prediction: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero samples from each signal.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.

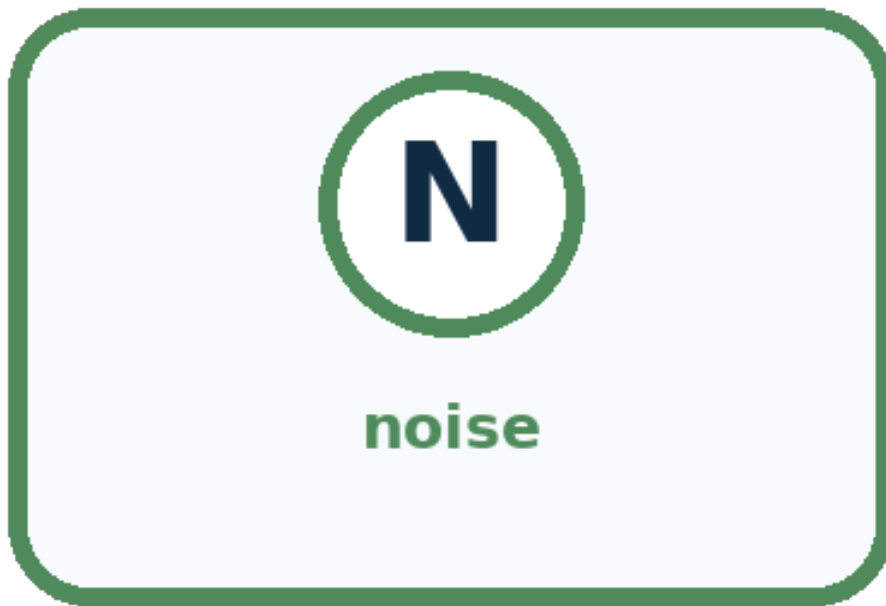




sample



amplitude



Word · Meaning · My example

Word: waveform

Meaning: graph of a signal changing over time

My example: _____

Word: sample

Meaning: one measurement at one moment

My example: _____

Word: amplitude

Meaning: size or strength of a signal value

My example: _____

Word: noise

Meaning: unwanted signal mixed with the target

My example: _____

Misconception detective

CLAIM TO REPAIR A loud sound must be a clap.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Signal

Meaning: A changing physical or recorded quantity that can carry information

Word: Sampling interval

Meaning: The time between successive recorded samples

Word: Transcript

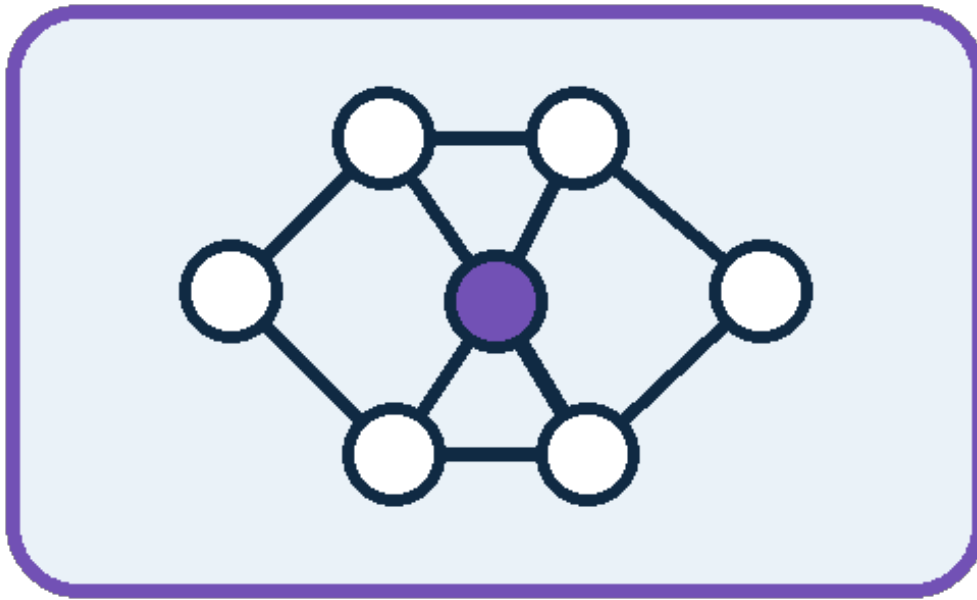
Meaning: A text representation of spoken words, which may contain errors or omit sound details

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a waveform.

STEP 1 — OBSERVE changing air pressure

STEP 2 — TRACE sample the signal and compare a clap pattern

STEP 3 — CONCLUDE clap detected, not detected, or uncertain

MODEL REASONING Look for two strong peaks separated by the expected gap. Noise may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when evidence is weak.

DO NOT OVERCLAIM A microphone measures a signal; it does not understand a speaker's meaning by itself.

What new evidence could change this answer?

A closer explanation

The largest value in 0,2,0,2,0 is 2. The largest value in 0,0,2,0,0 is also 2. The first sequence reaches 2 twice at different times, while the second does so once. Keeping only the maximum loses that distinction, just as counting black pixels lost their arrangement.

Sparse measurements can also miss changes. If we inspect the first sequence only at times 0, 2 and 4 milliseconds, all retained values are 0. The two intervening values of 2 disappear from the record. It would be incorrect to conclude from these retained samples alone that the signal stayed at zero between them.

More frequent sampling can preserve faster changes under suitable recording conditions, but sampling frequency alone does not make every recording clear. Microphone placement, background sound, clipping and the recording format can also matter. We are studying one source of missing information, not a complete audio engineering specification.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sound Becomes a Signal

A fictional microphone display records five samples during a short classroom tap.

The facts we will use

Times are 0,1,2,3,4 ms in order.

Signal A: 0,3,0,1,0. Signal B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

AI ACADEMY / AI-1/6

Same peak, different sample timing

	0 ms	1 ms	2 ms	3 ms	4 ms
Signal A	0	3	0	1	0
Signal B	0	1	0	3	0

Read a cell using both its row and column labels.

Sound Becomes a Signal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Timed sampling

Associate every measurement with its sampling time.



2. Signal A

Its largest sample 3 occurs at 1 ms.



3. Signal B

Its largest sample 3 occurs at 3 ms.



4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

CHANGE ONE THING

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Timed sampling

What happens and why: Associate every measurement with its sampling time.

Step: 2. Signal A

What happens and why: Its largest sample 3 occurs at 1 ms.

Step: 3. Signal B

What happens and why: Its largest sample 3 occurs at 3 ms.

Step: 4. Maximum summary

What happens and why: Both maxima are 3, but the strongest recorded tap arrives at different times.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling samples changes the recorded signal even if its maximum stays fixed.

What this example does not establish

These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: One clap

Evidence: single sharp peak

What it shows: target pattern

Case: Two claps

Evidence: two separated peaks

What it shows: different timing pattern

Case: Talking

Evidence: many irregular changes

What it shows: possible false trigger

Case: Quiet room

Evidence: small background values

What it shows: baseline

Repair two tempting claims

CLAIM A A loud sound must be a clap.

Repair:

CLAIM B Silence means the sensor has no data.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

A transcript might record the words Fine, thank you while omitting a pause, a laugh or overlapping speech. It might also record the wrong words. Text can be useful for searching spoken content, but it is not the same representation as the audio. Do not infer a speaker's emotions with certainty from a short transcript.

In a voice calculator, audio saying sixteen can be transcribed as sixty. A calculator can then correctly divide sixty by four and return fifteen. Inspecting the intermediate transcript identifies the first mismatch. Confirming the interpreted numbers before acting can prevent this particular error from travelling into a later stage.

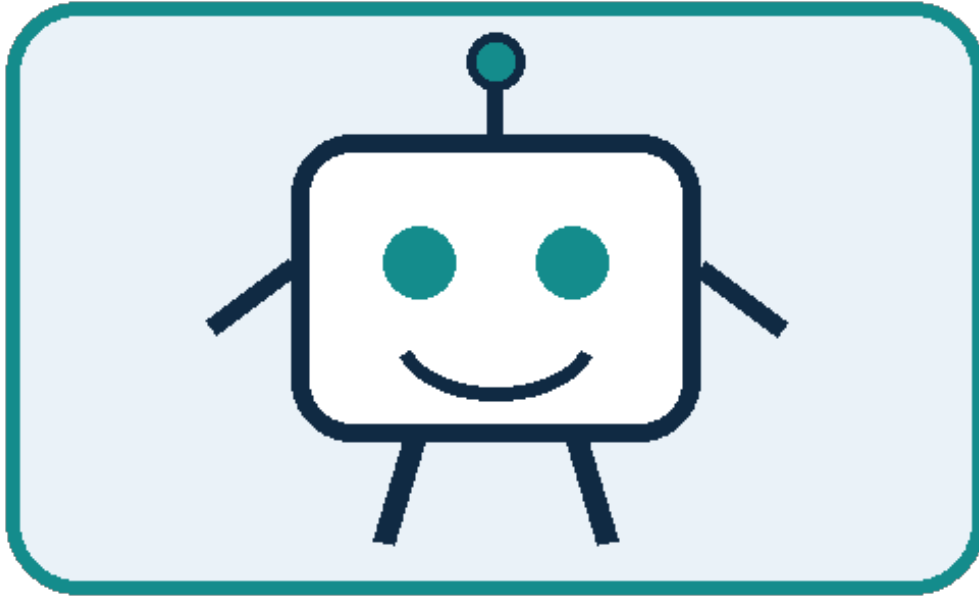
Explain one detail a transcript can preserve and one sound detail it may omit. Explain why a correct calculation does not prove the transcript was correct

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Computers receive changing measurements, not human-like hearing.

Use a fresh example to explain waveform.

Trace a new input → process → output example.

Find and repair the error in: Silence means the sensor has no data.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A five-sample record has values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. Read the timing before answering. The first-to-last duration is the final time minus the initial time, not the sample count multiplied by the gap.

State the interval and first-to-last duration. If only times 0,4,8 are kept, list the retained values and identify a change the shorter record misses

To test a claim about the spaces between samples, construct another possible sequence that agrees at retained times but differs between them. If both match the stored record, that record alone cannot decide which intervening pattern occurred.

Practise with less help: Inspect what was discarded

Use workbook W2 for the connected example “Sound as sampled information”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Sound as sampled information. Proposed explanation: “Five samples make five gaps”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does the all-zero retained record prove the full signal was silent? Use the picture.

My explanation using a pictured step and the given facts

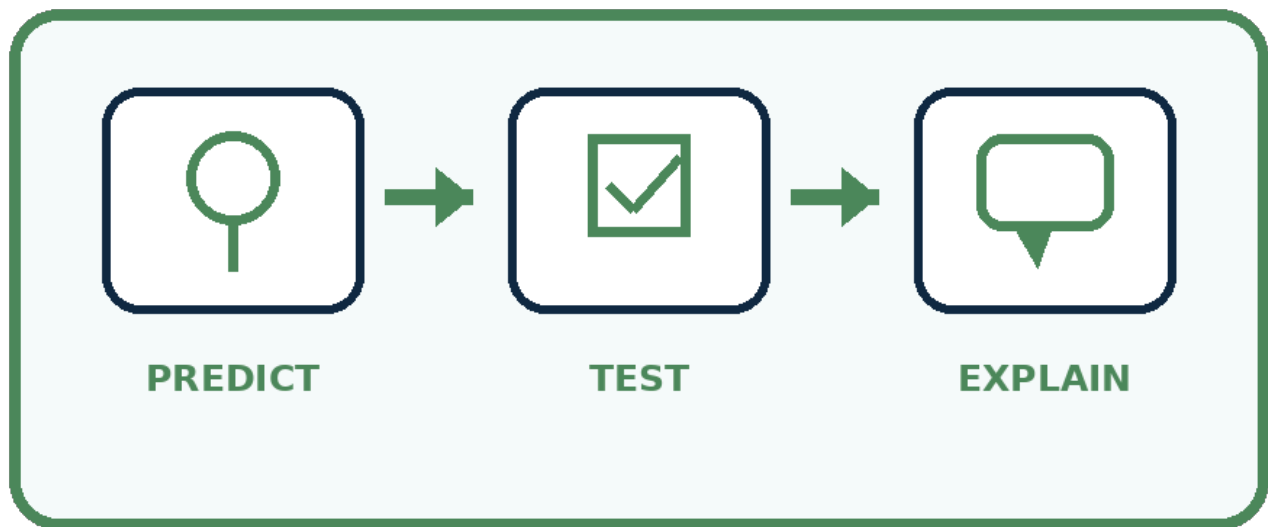
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Clap Pattern Detector

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Build a clap-pattern detector; test it with background noise.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

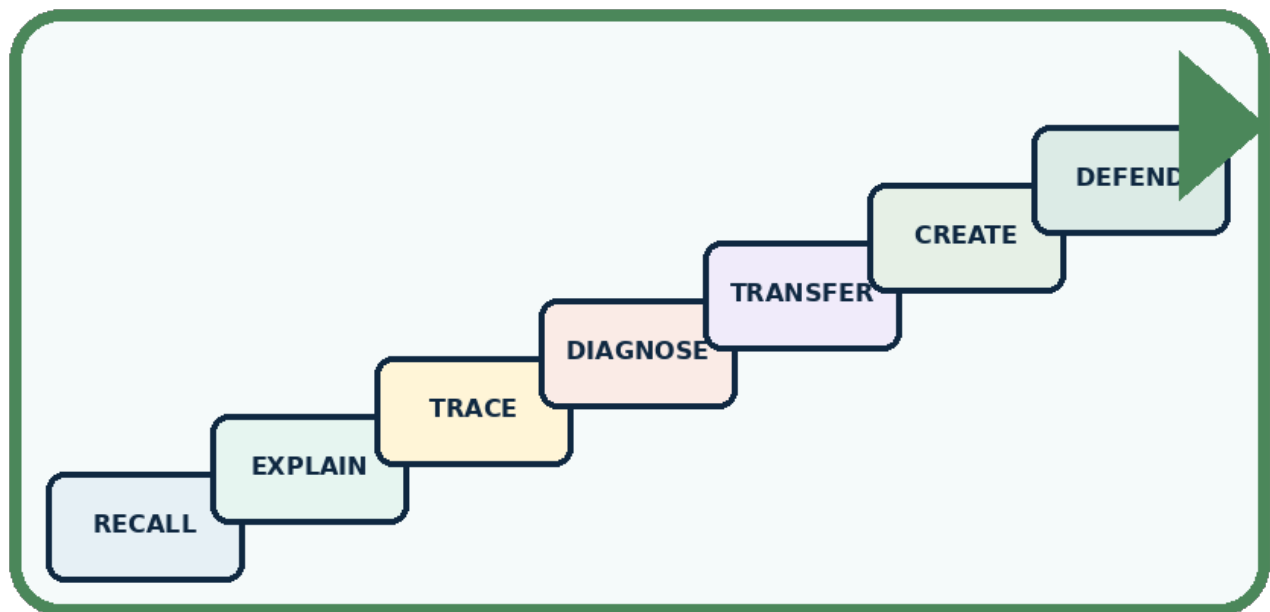
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define waveform.

EXPLAIN Computers receive changing measurements, not human-like hearing.

TRACE Show the input, process, and output.

DIAGNOSE Repair: A loud sound must be a clap.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

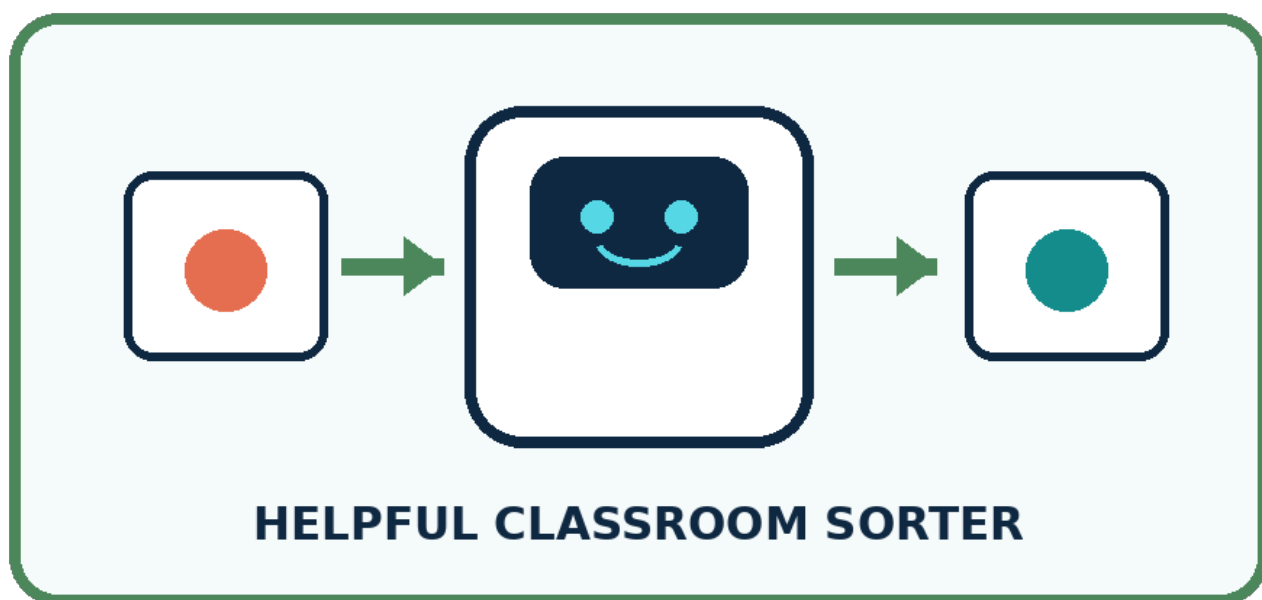
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT If the sorter uses sound, specify the exact sound pattern needed and test whether classroom noise makes that input unreliable.

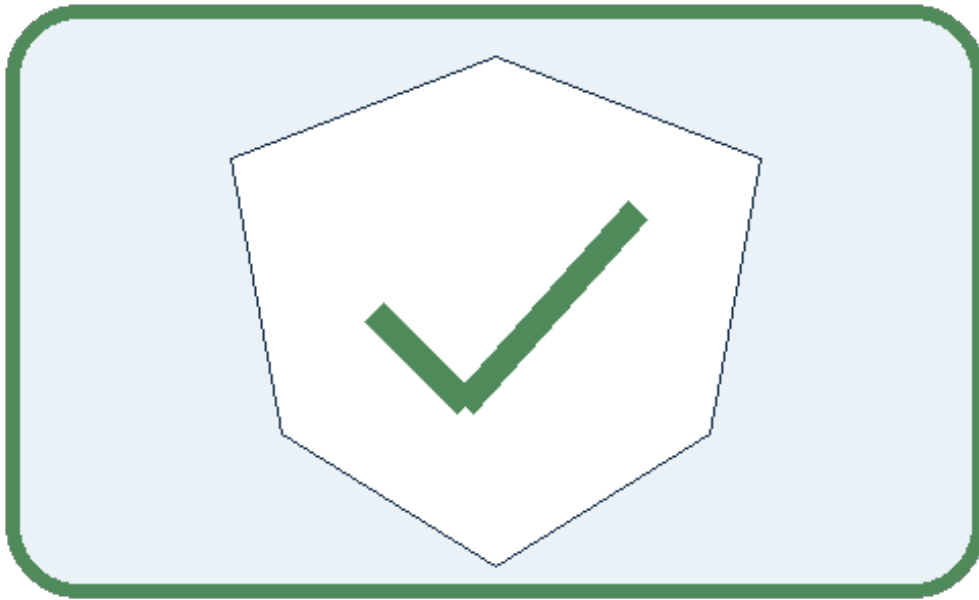
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN A microphone measures a signal; it does not understand a speaker's meaning by itself.

My project decision:

Evidence for your project

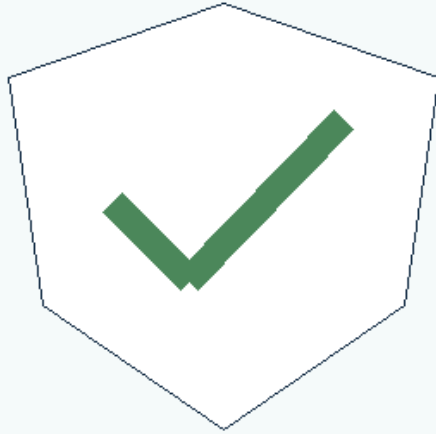
Record the sorter's sampling interval and retain relevant sequences when a decision depends on changes over time.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Use a waveform strip to decide whether a two-clap pattern occurred, then explain how background noise could cause a missed or false detection.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Look for two strong peaks separated by the expected gap. Noise may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when evidence is weak.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Categories and Labels.

13. Study a complete case

Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

A microphone records quiet and loud claps at the same distance and settings.

The question

What waveform difference is expected? Does it reveal a spoken word?

Worked explanation

The louder clap generally produces larger amplitude. Amplitude is not word meaning. Distance, gain and noise must be controlled.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Sound samples show changes in amplitude

Evidence or stage · Value or result

Evidence or stage: Quiet clap

Value or result: 1

Evidence or stage: Loud clap

Value or result: 3

A microphone records quiet and loud claps at the same distance and settings.

What waveform difference is expected? Does it reveal a spoken word?

The louder clap generally produces larger amplitude. Amplitude is not word meaning. Distance, gain and noise must be controlled.

14. Try a new case independently

Independent case: Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-sample sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times. Explain why the retained record cannot prove that nothing changed between measurements.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom noise makes that input unreliable.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A whisper recorded close to a microphone has larger sample amplitudes than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.